SIGNAL PROCESSING IN PRACTICE

Linear Prediction

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Linear Prediction

* Time series modeling
* Stationary processes

Applications

* Speech processing
* Financial data ✓
* Biomedical data (EEG, ECG, ...)

Origin

✓ "Theory of Linear Prediction" — P.P. Vaidhyanathan (Chap 2)

* Compression or coding application
    * Speech / audio (1970)
* Analysis of audio (speech recognition)
    * Biomedical data
* Forecasting ....

What is linear prediction

* $x[n]$ is w.s.s. sequence

LP coefficients

conjugate

past samples

forward pred.

$$\hat{x}_N^{(f)}[n] = - \sum_{i=1}^{N} a_{N,i}^{*} \left(x[n-i] \right)$$

order of pred.

[✓ Predict the current sample based on past samples]

✓ using a linear model

Error in prediction (error signal)

$$e_N^f[n] = x[n] - \hat{x}_N^f[n]$$

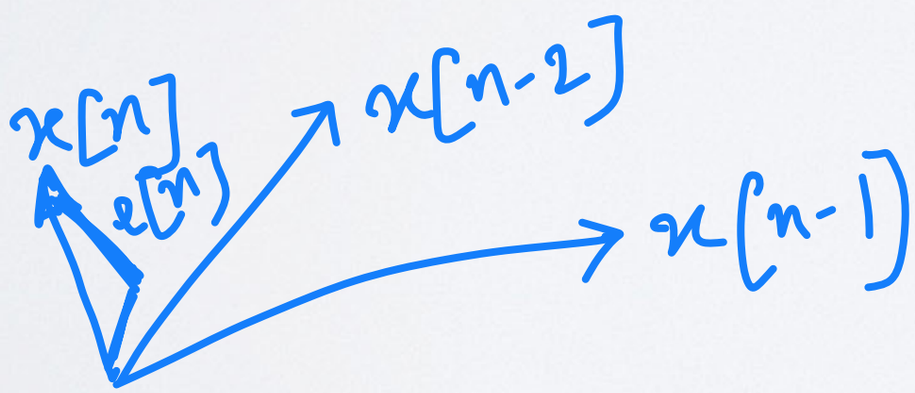
$$= x[n] + \sum_{i=1}^{N} a_{N,i}^* \, x[n-i]$$

[FIR filter]

$$x[n] \longrightarrow \boxed{A(z)} \longrightarrow e_N^f[n]$$

$$\mathcal{E}_N^f \triangleq E\left[|e_N^f[n]|^2 \right]$$

↓ minimize w.r.t. $\{a_{N,i}\}_{i=1}^{N}$



Linear Prediction Problem

- Consider a WSS $x[n]$
- The predicted output is given by $\hat{x}_N^f[n] = -\sum_{i=1}^{N} a_{N,i}^* x[n-i]$
- The error

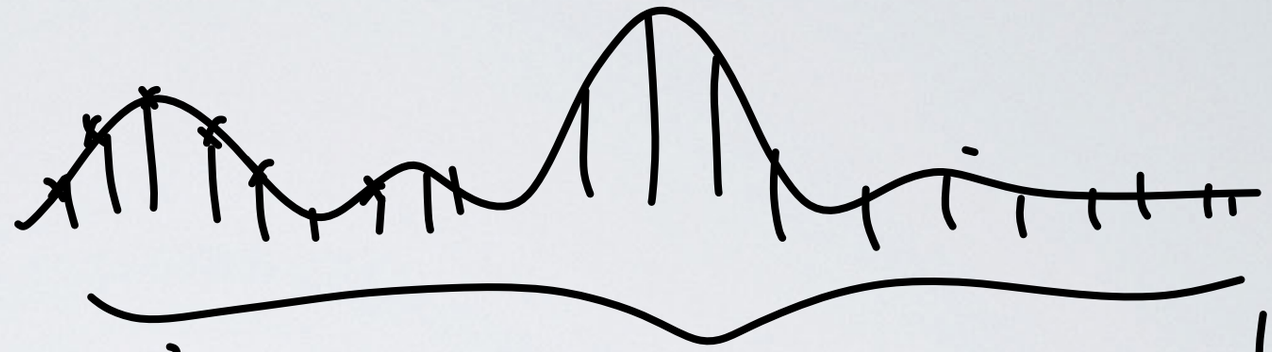
$$e_N^f[n] = x[n] - \hat{x}_N^f[n] = x[n] + \sum_{i=1}^{N} a_{N,i}^* x[n-i]$$

- The coefficients $\{a_{N,i}\}_{i=1}^{N}$ are given by min $E\{|e_N^f[n]|^2\}$
- Filtering analogy

Minimum phase

$$x[n] \rightarrow \boxed{A_N(z)} \rightarrow e_N^f[n] \qquad \hat{e}_N^f[n] \rightarrow \boxed{\frac{1}{A_N(z)}} \rightarrow x[n]$$

$x[n] =$



100 samples

1600

16 bits $\longrightarrow$ 10 bits

LP

a_N, i (10×16)

$\hat{e}_N^f$ $(100 \times 8 \text{ bits})$ Play

1000 bits

$\hat{\hat{x}}[n]$

$\hat{x}[n]$

$\dfrac{1}{A_N(z)}$ $\longrightarrow$ $\hat{x}[n]$

Orthogonality Principle

$\{e[n] \perp x[n-i]\}$ $\qquad E\left[e[n]\, x^*[n-i]\right] = 0$

Optimal linear predictor $\rightleftarrows$ $e_N[n] \perp x[n-i]$

$i = 1 \ldots N$

$\exists\ e_\perp$ which satisfies $E\left[e_\perp x_i^*\right]$; $\qquad x[n-i] = x_i$

$1 \le i \le N$

optimal

$e = x - \hat{x}$

$= x - \hat{x}_\perp + \hat{x}_\perp - \hat{x}$

$e_\perp = x - \hat{x}_\perp$

predicted value for orthogonal pred.

$\left\{ a^*_{N,1} \ \vdots \ a^*_{N,N} \right\} \longleftarrow \left\{ a^\perp_{N,1} \ \ a^\perp_{N,N} \right\}$

$$E\left[|e[n]|^2\right] = E\left[e[n]\, e^*[n]\right]$$

$$= E\left[\left|\underbrace{(x - \hat{x}_\perp)}_{e_\perp} + \underbrace{\hat{x}_\perp - \hat{x}}_{\text{linear comb. of past samples}}\right|^2\right]$$

$$= E\left[|e_\perp|^2\right] + \underbrace{E\left[|\hat{x}_\perp - \hat{x}|^2\right]}_{\geq 0}$$

Thus, if there exists a predictor, which generates an error signal $\perp \{x[n-i]\}_{i=1}^{N}$ $\rightarrow$ that is the optimal pred.

$$E[e[n]\,x^*[n-i]] = 0 \quad \forall\ i = 1 \dots N$$

$$\underline{a} = \begin{bmatrix} a_{N,1} \\ \vdots \\ a_{N,N} \end{bmatrix} \qquad \underline{x}[n] = \begin{bmatrix} x[n-1] \\ \vdots \\ x[n-N] \end{bmatrix}$$

$$e[n] = x[n] + \underline{a}^H \underline{x}$$

$$E[e[n]\,\underline{x}^H[n]] = \underline{0}$$

$$E[(x[n] + a^H \underline{x}[n])\,\underline{x}^H[n]] = \underline{0}$$

$$E\left[x[n]\,\underline{x}^H[n]\right] = -a^H E\left[\underline{x}[n]\,\underline{x}^H[n]\right]$$

$$= -a^H \mathbb{R}$$

γ^H ✓

$$E\left[\begin{bmatrix} x[n-1] \\ \vdots \\ x[n-N] \end{bmatrix} \begin{bmatrix} x^*[n-1] & \cdots & x^*[n-N] \end{bmatrix}\right]$$

$\mathbb{R}$

$$R(k) = E\left[x[n]\,x^*[n-k]\right]$$

$$\underline{\underline{R_N}} = \begin{bmatrix} R(0) & R(1) & \cdots & R(N-1) \\ R^*(1) & R(0) & \cdots & \\ \cdots & \cdots & \cdots & R(0) \end{bmatrix}$$

✓ ✓ ✓ ✓ ✓

Hermitian matrix

$$E\left[x[n]\,\underline{x}^H[n]\right] = E\left(x[n]\left[x^*[n-1] \;\cdots\; x^*[n-N]\right]\right)$$

$$= \left[R(1) \;\cdots\; \cdots \;R(N)\right]$$

$$\underline{\underline{r}} = \begin{bmatrix} R^*(1) \\ \vdots \\ R^*(N) \end{bmatrix} \qquad \left(\underline{r}^H = -a^H \mathbb{R}_N\right)^H$$

$$\mathbb{R}\,\underline{a} = -\underline{r}$$

$$\boxed{\underline{a} = -\mathbb{R}_N^{-1}\,\underline{r}}$$

Yule Walker Equations

Wiener Hopf equations

Solving Linear Prediction Problem

- Orthogonality Principle - optimal LP model satisfies
  - Error signal $e_N^f[n] \perp \{x[n-i]\}_{i=1}^N$ which means

$$E\{e_N^f[n]x^*[n-i]\} = 0$$

$a = -R^{-1} \underline{r}$

- Using the above set of equations $\boldsymbol{R}\boldsymbol{a} = \boldsymbol{r}$

$$\boldsymbol{R} = \begin{bmatrix} R(0) & R(1) & \dots & R(N-1) \\ R^*(1) & R(0) & \dots & R(N-2) \\ \dots & \dots & \dots & \dots \\ R^*(N-1) & R^*(N-2) & \dots & R(0) \end{bmatrix} \quad \boldsymbol{a} = \begin{bmatrix} a_{N,1} \\ a_{N,2} \\ \dots \\ a_{N,N} \end{bmatrix} \quad \boldsymbol{r} = -\begin{bmatrix} R^*(1) \\ R^*(2) \\ \dots \\ R^*(N) \end{bmatrix}$$

- Normal/Yule-walker/Weiner-Hopf Equations

Optimal value of energy of error signal $a^*_{N,i}$

$$E\left[\left|e^{f*}_N[n]\right|^2\right] = E\left[\underbrace{e[n]}\, e^*[n]\right]$$

$$\left\{\underbrace{x^*[n]} + \sum_{i=1}^{N} a_i \underbrace{x^*[n-i]}\right\}$$

$$= E\left[\left(\underbrace{x[n]} + \sum_{i=1}^{N} a_i^* x[n-i]\right) \underbrace{x^*[n]}\right]$$

$$E^{*f}_N = \underbrace{R(0) + \sum_{i=1}^{N} a_i^* R^*(i)}_{= R(0) + \underline{a}^H \underline{r}} \quad \checkmark$$

Estimating Autocorrelations

$$R(k) = E[x(n)\,x^*(n-k)]$$

$x[n]$ of length L

$$x_L[n] = \begin{cases} x[n] & 0 \le n \le L-1 \\ 0 & \text{outside} \end{cases}$$

I

$$\hat{R}(k) = \frac{1}{L} \sum_{n=0}^{L-1} x_L[n]\, x_L^*[n-k) \quad ; \quad k = 0, \ldots, N$$

Different # sum terms for different $-k$

II

$$\hat{R}(k) = \frac{1}{L} \sum_{n=0}^{L-1-k} x[n]\, x^*[n-k]$$

$x[0] \quad x[1] \quad x[2] \quad x[3] \quad x[4] \qquad \left(\begin{matrix} L=5 \\ N=3 \end{matrix}\right)$

$\mathbb{R}_3$ $\quad$ I

$$X = \begin{bmatrix} x[0] & 0 & 0 \\ x[1] & x[0] & 0 \\ x[2] & x[1] & x[0] \\ x[3] & x[2] & x[1] \\ x[4] & x[3] & x[2] \\ 0 & x[4] & x[3] \\ 0 & 0 & x[4] \end{bmatrix}$$

$$\hat{\mathbb{R}}_3 = \frac{1}{L} X^H X$$

II

$$X = \begin{bmatrix} x(2) & x(1) & x(0) \\ x(3) & x(2) & x(1) \\ x(4) & x(3) & x(2 \end{bmatrix}$$

$$\hat{\mathbb{R}}_3 = \frac{1}{L} X^H X$$

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$$\mathbb{R}\,\underline{a} = -\underline{r} \qquad \underline{a} = \begin{bmatrix} a_{N,1} \\ \vdots \\ a_{N,N} \end{bmatrix}$$

$$R(0) + \underline{r}^H \underline{a} = \underline{\underline{\epsilon_N^f}}$$

$$R(0) + \underline{r}^T \underline{a} = \epsilon_N^f \qquad \underline{\underline{\underline{r}}} = \begin{bmatrix} R^*(1) \\ \vdots \\ R^*(N) \end{bmatrix}$$

$$\underline{r} + \mathbb{R}\,\underline{a} = 0$$

$$\begin{bmatrix} R(0) & R(1) & \cdots & R(N) \\ R^*(1) & R(0) & \cdots & \\ \vdots & & & \\ R^*(N) & R^*(N-1) & \cdots & R(0) \end{bmatrix} \begin{bmatrix} 1 \\ a_{N,1} \\ \vdots \\ a_{N,N} \end{bmatrix} = \begin{bmatrix} \epsilon_N^f \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\mathbb{R}_{N+1}\,\underline{a}_N = \begin{bmatrix} \epsilon_N^f \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\mathbb{R}$

$$\begin{bmatrix} R(0) & R(1) & \cdots & R(N) \\ \vdots & & & \vdots \\ R^*(N) & & \cdots & R(0) \end{bmatrix}_{N+1 \times N+1} \qquad \begin{bmatrix} 1 \\ a_{N,1} \\ \vdots \\ a_{N,N} \end{bmatrix}$$

$$\underline{a}_N^H\,\mathbb{R}_{N+1}\,\underline{a}_N = \underline{\underline{\epsilon_N^f}}$$

positive semi-definite, Toeplitz

linear pred.

$\{ x_{w_1}[1] \dots x_{w_1}[L] \}$

$\{ x_{w_{10}}[1] \dots x_{w_{10}}[L] \}$

STLP

N^{th} order predictor $\quad \underline{a} = -\mathbb{R}^{-1}\underline{r} \qquad N = 20$

order of computations $\sim \underline{\underline{O(N^3)}}$

Levinsion Durbin Recursion ✓

3rd order

$$\mathbb{R}_4 \, \underline{a}_3 = \begin{bmatrix} E_3^f \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad R_4 = \begin{bmatrix} R(0) & R(1) & R(2) & R(3) \\ R^*(1) & R(0) & R(1) & R(2) \\ R^*(3) & R^*(2) & R^*(1) & R(0) \end{bmatrix}$$

$$\underline{a}_3 = \begin{bmatrix} 1 \\ a_{31} \\ a_{32} \\ a_{33} \end{bmatrix}$$

$$\underbrace{\overset{\mathbb{R}_4}{\begin{bmatrix} R(0) & R(1) & R(2) & R(3) & R(4) \\ R^*(1) & R(0) & R(1) & R(2) & R(3) \\ & & & & R(2) \\ R^*(3) & R^*(2) & R^*(1) & R(0) & R(1) \\ R^*(4) & R^*(3) & \cdots & R^*(1) & R(0) \end{bmatrix}}}_{\mathbb{R}_5} \begin{bmatrix} 1 \\ a_{3,1} \\ a_{3,2} \\ a_{3,3} \\ 0 \end{bmatrix} = \begin{bmatrix} \epsilon_3^t \\ 0 \\ 0 \\ 0 \\ \alpha_3 \end{bmatrix}$$

$$\alpha_3 \triangleq R^*(4) + R^*(3)\,a_{3,1} + R^*(2)\,a_{3,2} + R^*(1)\,a_{3,3}$$

$$\mathbb{R}_5\, \underline{a}_4 = \begin{bmatrix} \epsilon_4^t \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \overbrace{\begin{matrix} R(0) & R(1) & R(2) & R(3) \\ R^*(1) & R(0) & R(1) & R(2) \\ & & & \\ R^*(3) & R^*(2) & R^*(1) & R(0) \end{matrix}}^{\mathbb{R}_4} & \begin{matrix} R(4) \\ R(3) \\ R(2) \\ R(1) \end{matrix} \\ R^*(4) \quad R^*(3) \quad \cdots\cdots \quad R^*(1) & R(0) \end{bmatrix} \begin{bmatrix} 1 \\ a_{3,1} \\ a_{3,2} \\ a_{3,3} \\ 0 \end{bmatrix} = \begin{bmatrix} \epsilon_3^f \\ 0 \\ 0 \\ 0 \\ \alpha_3 \end{bmatrix} \quad \text{I}$$

Reverse the order of the rows, reverse order of columns, apply conjugate

$$\begin{bmatrix} \overbrace{\begin{matrix} R(0) & R(1) & R(2) & R(3) \\ R^*(1) & R(0) & R(1) & R(2) \\ & & & \\ R^*(3) & R^*(2) & R^*(1) & R(0) \end{matrix}}^{\mathbb{R}_4} & \begin{matrix} R(4) \\ R(3) \\ R(2) \\ R(1) \end{matrix} \\ R^*(4) \quad R^*(3) \quad \cdots\cdots \quad R^*(1) & R(0) \end{bmatrix} \begin{bmatrix} 0 \\ a_{3,3}^* \\ a_{3,2}^* \\ a_{3,1}^* \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_3^* \\ 0 \\ 0 \\ 0 \\ \epsilon_3^f \end{bmatrix} \quad \text{II}$$

Define constant $\underline{\underline{k_4^*}} = \dfrac{-\alpha_3}{\epsilon_3^f}$ $\alpha_3 = -k_4$

$$Eq(\mathrm{I}) + k_4^* \, Eq(\mathrm{II})$$

$$\Rightarrow \mathbb{R}_5 \left(\begin{bmatrix} 1 \\ a_{31} \\ a_{32} \\ a_{33} \\ 0 \end{bmatrix} + k_4^* \begin{bmatrix} 0 \\ a_{33}^* \\ a_{32}^* \\ a_{3,1}^* \\ 1 \end{bmatrix} \right) = \begin{bmatrix} * \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

ϵ_4^f

$$\epsilon_4^f = \epsilon_3^f + k_4^* \alpha_3^* = \epsilon_3^f + k_4^* \left(-\epsilon_3^f k_4 \right)$$

$$= \underline{\underline{\epsilon_3^f \left(1 - |k_4|^2 \right)}}$$

$$a_{4,1} = a_{31} + k_4^* a_{33}^*$$

$$a_{4,2} = a_{32} + k_4^* a_{32}$$

$$a_{43} = a_{33} + k_4^* a_{31} \qquad a_{44} = k_4^*$$

Summary

Let $A_m(z) = 1 + \underbrace{a^*_{m,1} z^{-1} + \cdots\cdots + a^*_{m,m} z^{-m}}$

$$\tilde{A}_m(z) = \underbrace{A^*(1/z)}$$

$$z^{-m}\tilde{A}_m(z) = a_{m,m} + a_{m,m-1} z^{-1} + \cdots\cdots + z^{-m}$$

L.D we just proved is

$$* \quad A_4(z) = A_3(z) + k_4 z^{-1}\left[z^{-3}\tilde{A}_3(z)\right]$$

Recursion

$m \to m+1$

m be current order

$$\alpha_m = R^*(m+1) + a_{m,1} R^*(m) + \dots + a_{m,m} R^*(1)$$

$$k_{m+1} = \frac{-\alpha_m^*}{\epsilon_m^f}$$

ϵ_m^f

$a_{m,1}$

$\vdots$

$a_{m,m}$

$$\epsilon_{m+1}^f = \left(1 - |k_{m+1}|^2\right) \epsilon_m^f$$

$$A_{m+1}(z) = A_m(z) + k_{m+1} z^{-1} \left[z^{-m} \tilde{A}_m(z) \right]$$

Initialization

$A_0(z) = 1$ $\quad e_0[n] = x[n]$ $\quad \epsilon_0^f = R(0)$

$$\epsilon_0^f = E\left[|e_0^f[n]|^2\right]$$

$$\alpha_0 = R^*(1) \Rightarrow k_1 = -\frac{\alpha_0^*}{\epsilon_0^f} = -\frac{R(1)}{R(0)}$$

$$\epsilon_1^f = \left(1 - \left|\frac{R(1)}{R(0)}\right|^2\right) R(0)$$

$$A_1(z) = 1 + k_1 z^{-1}(1) = \underline{\underline{1 + k_1 z^{-1}}}$$

$$\epsilon_1^f < \epsilon_0^f$$

$m^{-1} \to m$

<u>Complexity</u> $\approx m$ (multiplications) m (additions)

$O(N^2)$

Properties

- Minimum phase property for $A_N(z)$
- Autocorrelation matrix is Hermitian positive semi-definite, Toeplitz
- Min. err $\epsilon_N^f = E\{|e_N^f[n]|^2\} = R(0) + \sum_{i=1}^{N} a_{N,i}^* R^*(i)$
- Using the above expression, we get augmented normal equations.

$$\boldsymbol{R}_{N+1}\boldsymbol{a}_{N+1} = \begin{bmatrix} \epsilon_N^f \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

- One-to-one mapping between autocorrelation sequence and the linear prediction coefficients and the min. error.

Chap 5

A.R. process (autoregressive process)

A W.S.S. $w[n]$ is called AR(N) if

$$w[n] = -\sum_{i=1}^{N} d_i^* w[n-i] + e[n]$$

(i) $e[n]$ has zero mean and it is white

$$E[e[n]\, e^*[n-k]] = 0 \quad \forall \; k \neq 0$$

(ii) $D(z) = 1 + \sum_{i=1}^{N} d_i^* z^{-i}$ has all zeros inside unit circle

Linear prediction on a general sequence $x[n]$

$\downarrow$

$$E[e[n]\, x^*[n-i]] = 0$$

$\forall\, i = 1 \ldots N$

colored

An autoregressive AR(N) process

$x[n]$

$\downarrow$

$$e[n] \perp \underline{\underline{x[n-i]}}$$

$\forall\, i$

$\underline{\underline{\text{white}}}$

$x[n]$ is AR(N) process, then L.P. reveals.

$x[n] \xrightarrow{\text{approx.}}$ AR(N)

$x[n] \longrightarrow$ L.P. $\longrightarrow \boxed{A_N(z)} \longrightarrow e[n]$

not white

var ϵ_N^+

$\hat{e}[n]$ to some white noise

$\hat{e}[n] \longrightarrow \boxed{1/A_N(z)} \longrightarrow \hat{x}[n]$

white noise ϵ_N^+

$\sigma_e \left|H_N(e^{j\omega})\right|^2 = \hat{S}_{xx}(e^{j\omega}) \approx S_{xx}(e^{j\omega})$

A.R.

$$\underbrace{e[n]}_{\text{white}} \longrightarrow \boxed{1/A_N(z)} \longrightarrow x[n]$$

$$H_N(z) = 1/A_N(z)$$

$$S_{xx}(e^{j\omega})$$

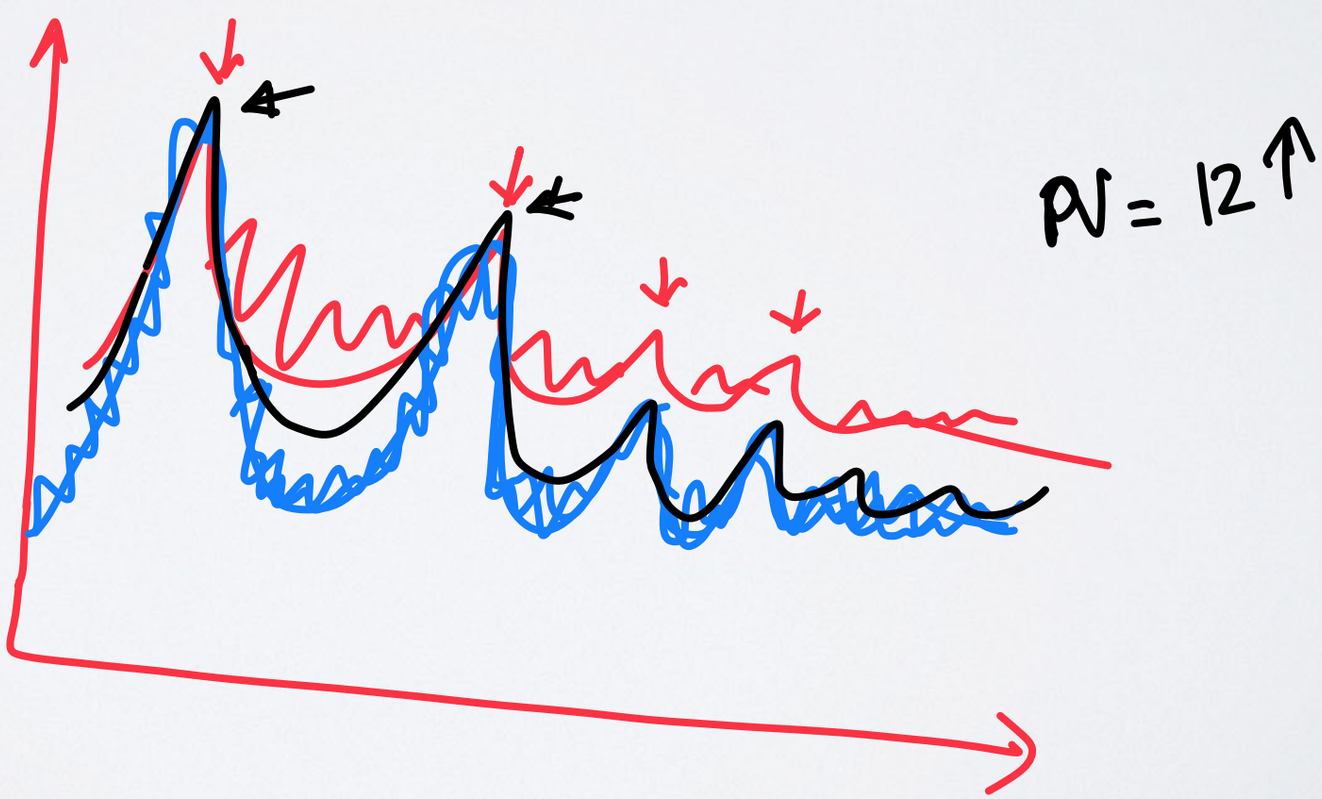
Power spectrum

$$S_{ee}(e^{j\omega})$$

$$S_{xx}(e^{j\omega}) = |H_N(e^{j\omega})|^2 \underbrace{S_{ee}(e^{j\omega})}_{\text{constant}}$$

$$S_{xx}(e^{j\omega}) = \sigma_e^2 \, |H_N(e^{j\omega})|^2$$

$$\underline{\underline{H_N(z)}} = \frac{1}{1 + a_{N,1} z^{-1} + \cdots + a_{N,N} z^{-N}}$$

$$= \frac{1}{\prod_{k=1}^{N} (1 - p_k z^{-1})}$$

$$\underline{\underline{|H_N(e^{j\omega})|^2}}$$